

Operating System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Suppose that a disk has 100 cylinders, numbered 0 to 99. The drive is currently serving a request at cylinder 45. The queue of pending request, in FIFO order is: 88, 72, 13, 74, 48, 9, 22, 50, 35 and 50. What is the total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of the following disk scheduling algorithms? a) FCFS b) SCAN c) SSTF

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Why virtual memory technique is used in the computer system? What is logical address? Explain the process of conversion of logical address to physical address in single level paging scheme.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is critical section? What are the operations that can be performed on the semaphore? Explain the solution to producer consumer problem using semaphore.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Draw and describe the 5-state process model.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is kernel? Differentiate between monolithic and microkernel structure.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is DMA? Explain how it works in brief with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What are different types of file? Describe the sequential and random access of files.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is deadlock? What are necessary conditions for deadlock? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is thread? Explain user level thread and kernel level thread.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. A 2 GB hard disk has 2 KB block size, calculate the size of bitmap for bitmap based free space management.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is an operating system? Differentiate between time sharing and real time operating system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Consider the following set of processes, with the length of CPU burst time given in milliseconds.

Process	CPU Burst (ms)	Priority	Arrival Time (ms)
A	83	0 (Lowest)	0
B	91	1	10
C	20	2	20
D	64	3 (Highest)	0

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.